Global Fuel Prices and Taxation Analysis

An Interactive Dashboard Study of 84 Countries, 2020-2026

Capstone Project: Data Analysis and Visualization (Tableau)

Submitted by: Vidhya Jain
Enrollment no - EN25CS3011044

https://public.tableau.com/app/profile/vidhya.jain/viz/GlobalFuelPricesAnalysis/Story1?publish=yes

1. Abstract

This report studies global fuel prices and taxation from January 2020 to April 2026, using 27,468 weekly records for 84 countries in 7 regions. The data covers petrol, diesel and LPG prices in US dollars per litre, the Brent crude oil price, the tax percentage on fuel, and each country's income level and subsidy level. The data was cleaned and enriched in Python, then explored through 11 charts and a three-view interactive dashboard built for three user personas: an energy policy consultant, a logistics supply chain manager and an energy economics researcher.

The main finding is that subsidy policy sorts countries into price tiers more than any other variable. Countries with Low subsidies average 3.45 USD/L for petrol, while Very High subsidy countries average 0.15 USD/L. Every one of the 44 low-subsidy countries taxes fuel at roughly 37%, while the other 40 countries sit near 15%. Within any single country, pump prices follow Brent crude closely (average correlation 0.95), and the global average petrol price rose about 78% between 2020 and 2026.

2. Introduction

2.1 Background

Fuel prices affect household budgets, business costs and government revenue. Governments shape them through taxes and subsidies, while the world oil price, represented here by Brent crude, sets the underlying cost. The same crude oil price can produce very different pump prices depending on policy, which makes cross-country comparison valuable for policy makers, businesses and researchers.

2.2 Objectives

- Compare petrol, diesel and LPG prices across regions, income levels and subsidy levels.
- Measure how taxation differs across regions and subsidy groups.
- Track price trends over time and their relationship with Brent crude.
- Identify low-cost and high-volatility markets for fuel procurement.
- Deliver an interactive dashboard that answers the questions of three user personas.

2.3 User scenarios

Persona	Need
Dr. Sarah Mitchell, Energy Policy Consultant	Compare tax and subsidy policies across regions to support fair and sustainable fuel taxation.
Marcus Thompson, Global Supply Chain Manager	Find regions with favourable fuel prices, understand volatility and long-term trends for procurement.
Prof. Elena Rodriguez, Energy Economics Researcher	Study how income level relates to pricing, taxation and subsidies, and the effect on affordability.

3. Dataset Description

The dataset (global_fuel_prices_Dataset.xlsx) holds 27,468 rows and 10 columns. Each of the 84 countries has exactly 327 weekly observations from 6 January 2020 to 6 April 2026. There are no missing values and no duplicate country-week rows.

Column	Type	Description
date	Date	Week start date
country	Text	Country name (84 values)
region	Text	Africa, Asia, Europe, Middle East, North America, Oceania, South America
income_level	Text	Low, Middle or High
subsidy_level	Text	Low, Medium, High or Very High
petrol_usd_liter	Number	Petrol price, USD per litre
diesel_usd_liter	Number	Diesel price, USD per litre
lpg_usd_liter	Number	LPG price, USD per litre
brent_crude_usd	Number	Brent crude oil price, USD
tax_percentage	Number	Tax percentage applied to fuel

4. Data Preprocessing

Preprocessing was done in Python with pandas (prepare_data.py). The data quality checks confirmed complete data, so the work focused on making the file easier to analyze in Tableau:

- Date parts: year, quarter and month columns were added for time-based charts.
- Map-friendly names: UAE became United Arab Emirates and Ivory Coast became Cote d'Ivoire, so Tableau's geographic roles recognize all 84 countries.
- Derived measures: diesel-petrol gap, tax amount in USD per litre of petrol, pre-tax petrol price, and petrol-to-Brent ratio.
- Summary tables: country-level, region-by-year and income-by-subsidy summaries saved as CSV files.

Data observations. Four properties of the data shape how it should be interpreted. (1) Petrol, diesel and LPG prices move almost identically (correlation about 0.999), so conclusions for petrol hold for the other two fuels. (2) Brent crude has a single global value each week. (3) Weekly tax_percentage varies widely within a country (average standard deviation 12.4 points) and is uncorrelated with Brent, so country averages are more reliable than individual weeks. (4) Pooled across countries, petrol and Brent correlate only 0.26 because price levels differ greatly between countries; inside one country the correlation is 0.95.

5. Tools and Methodology

- Python (pandas, matplotlib) for cleaning, summary statistics and report charts.
- Tableau Desktop / Tableau Public as the target business intelligence tool, with the cleaned CSV as data source.

- A self-contained HTML dashboard (Chart.js) that mirrors the three Tableau dashboard views and runs in any browser.

Analysis followed four steps: profile the data, aggregate by region, income level, subsidy level and country, compare trends over time against Brent, and measure volatility using the coefficient of variation (standard deviation divided by mean) of weekly petrol prices.

6. Analysis and Findings

6.1 Scenario 1: Policy view for Dr. Mitchell

Average petrol prices differ by a factor of three between regions: Europe (3.70 USD/L) and Oceania (3.66) are the most expensive, while the Middle East (1.23) and South America (1.43) are the cheapest.

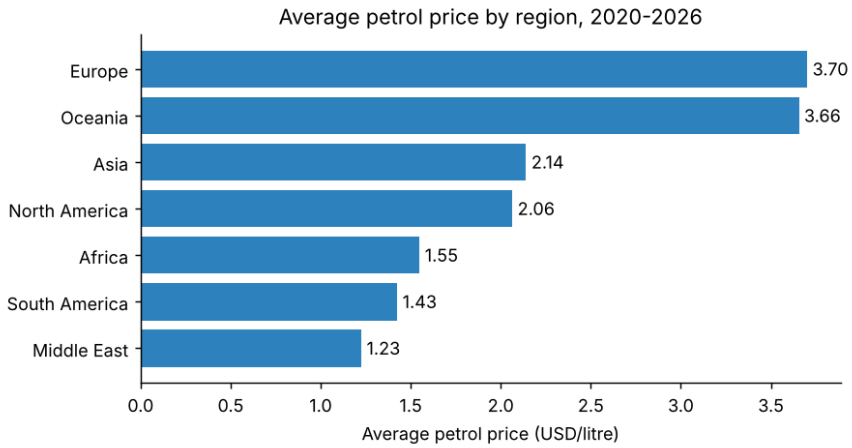


Figure 1: Average petrol price by region

Taxation follows a similar order. Oceania (38.7%) and Europe (34.6%) tax fuel most heavily, while South America (18.2%) and the Middle East (19.6%) tax it least.

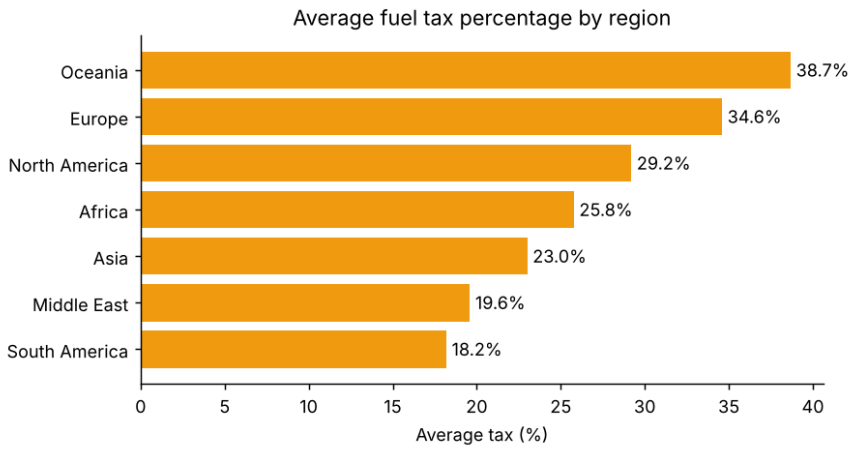


Figure 2: Average tax percentage by region

Subsidy level is the strongest single driver of price. Moving from Low to Very High subsidy lowers the average petrol price from 3.45 to 0.15 USD/L, a drop of about 96%.

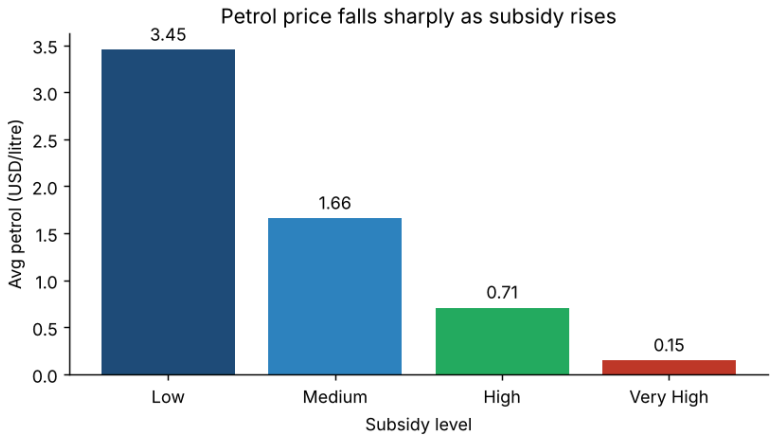


Figure 3: Average petrol price by subsidy level

Subsidy level	Petrol USD/L	Diesel USD/L	LPG USD/L	Avg tax %
Low	3.45	3.24	2.59	37.5
Medium	1.66	1.52	1.24	14.8
High	0.71	0.65	0.53	15.1
Very High	0.15	0.13	0.11	14.9

Tax clusters. Plotting each country's average tax against its average petrol price shows two distinct groups. All 44 Low-subsidy countries average about 37% tax, and all 40 countries in the other three subsidy tiers average about 15%. Tax policy and subsidy policy therefore move together, and a policy maker comparing tax rates should compare countries within the same subsidy tier.

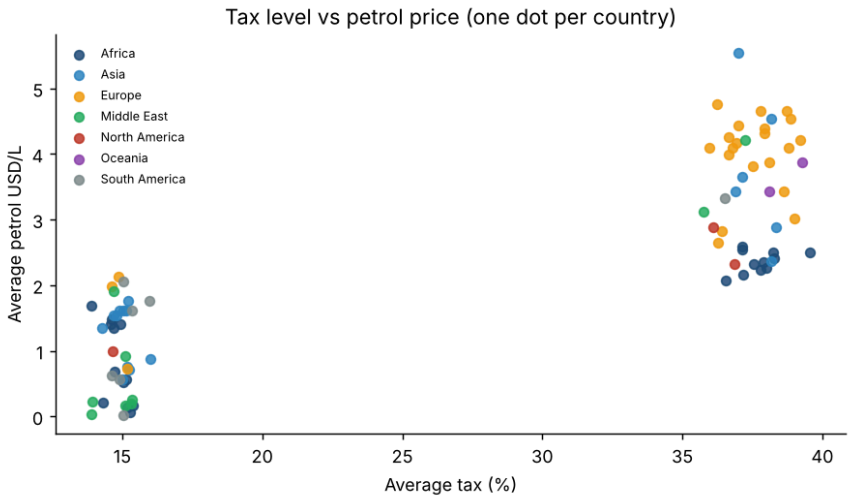


Figure 4: Average tax versus average petrol price, one dot per country

6.2 Scenario 2: Logistics view for Mr. Thompson

All ten of the cheapest petrol markets are Very High-subsidy countries, mostly oil producers: Venezuela (0.02 USD/L), Iran (0.04), Libya (0.06), Algeria (0.14), Iraq (0.16), Egypt (0.17), Kuwait (0.19) and Nigeria (0.21). The most expensive include Hong Kong (5.54), Norway (4.76), the Netherlands (4.66), Ireland (4.65), Denmark and Singapore (both 4.54).

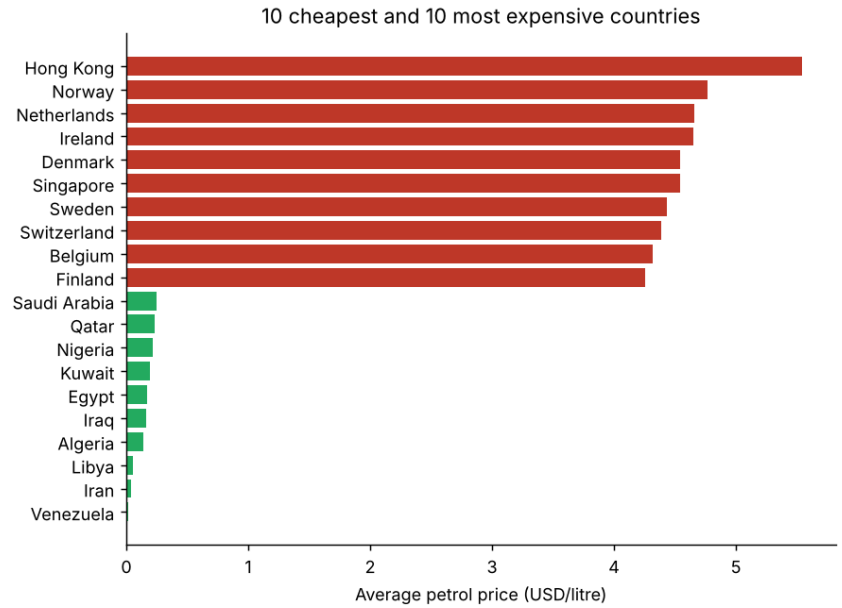


Figure 5: The 10 cheapest and 10 most expensive countries by average petrol price

The global average petrol price rose from 1.49 USD/L in 2020 to 2.28 in 2021, reached 2.55 in 2024 and stood at 2.65 in early 2026, about 78% above 2020. Brent crude moved from an average of 77.6 USD in 2020 to 123.7 USD in 2026 (about +60%). Every region followed the same pattern: a dip in spring 2020, a sharp rise to a peak in 2022, a fall in early 2023 and a slow climb since.

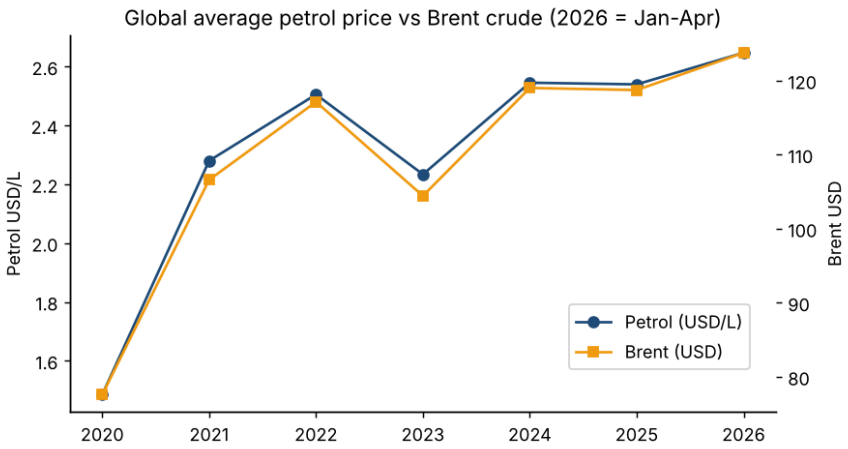


Figure 6: Global average petrol price and Brent crude by year

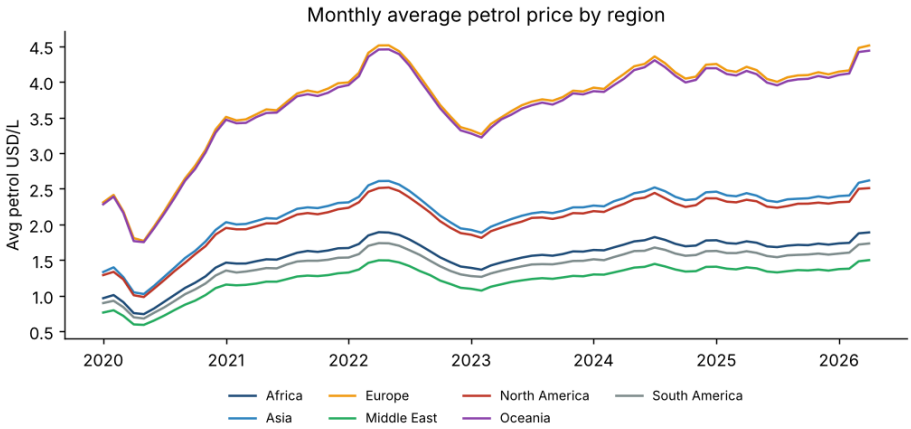


Figure 7: Monthly average petrol price by region

Volatility differs sharply by market. Venezuela (coefficient of variation 0.60), Iran (0.50) and Libya (0.39) are the most volatile, followed by Algeria, Iraq and Nigeria; all six are Very High-subsidy countries. 75 of the 84 countries sit below 0.20, near 0.18. Cheap but volatile markets are risky to build procurement plans around. Diesel is cheaper than petrol in 95.5% of weekly records, which favours diesel-fuelled fleets.

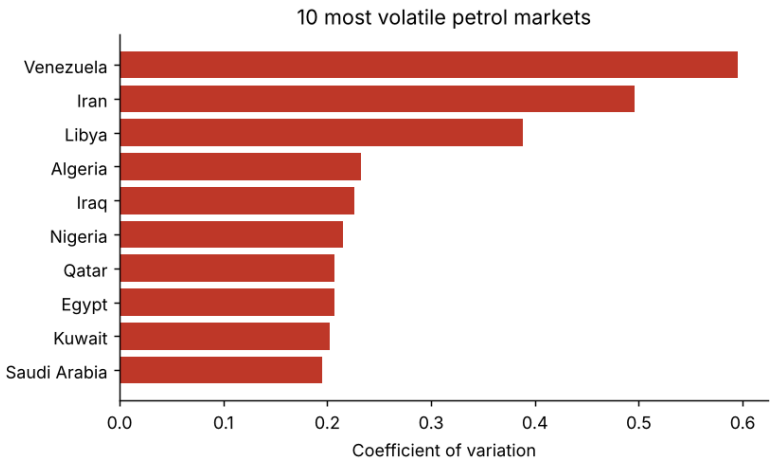


Figure 8: Ten most volatile petrol markets

6.3 Scenario 3: Research view for Prof. Rodriguez

Income level and price are linked mainly through subsidy policy. High-income countries pay an average of 3.55 USD/L and tax at 34.7%, compared with 1.59 USD/L and 21.1% for middle-income countries and 1.39 USD/L and 22.9% for low-income countries. Of the 32 high-income countries, 28 are Low-subsidy; only 4 offer High or Very High subsidies.

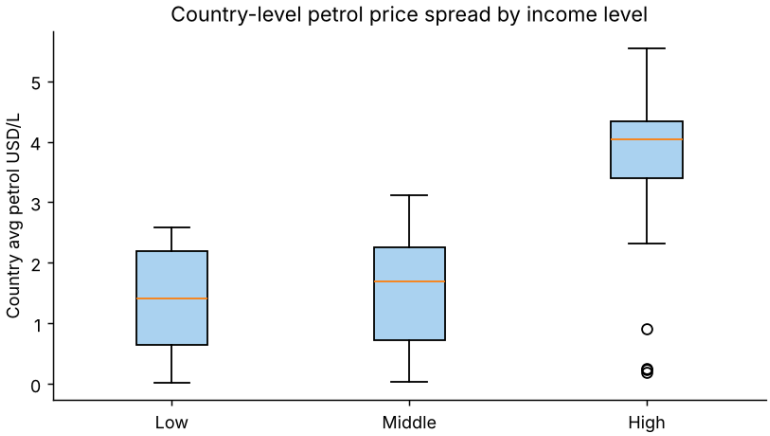


Figure 9: Country-level petrol price spread by income level

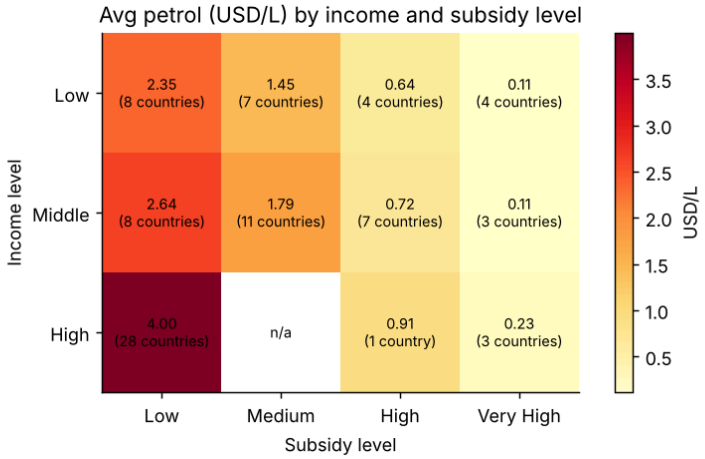


Figure 10: Average petrol price by income and subsidy level

Low- and middle-income countries are spread across all four subsidy tiers. In the same subsidy tier, income makes a small difference: Low-subsidy middle-income countries pay 2.64 USD/L, Low-subsidy low-income countries 2.35, and Low-subsidy high-income countries 4.00. Very High -subsidy countries pay about 0.11 USD/L in low and middle-income groups. Fuel is therefore most affordable where subsidies are largest, though this carries a fiscal cost for those governments, especially when Brent rises.

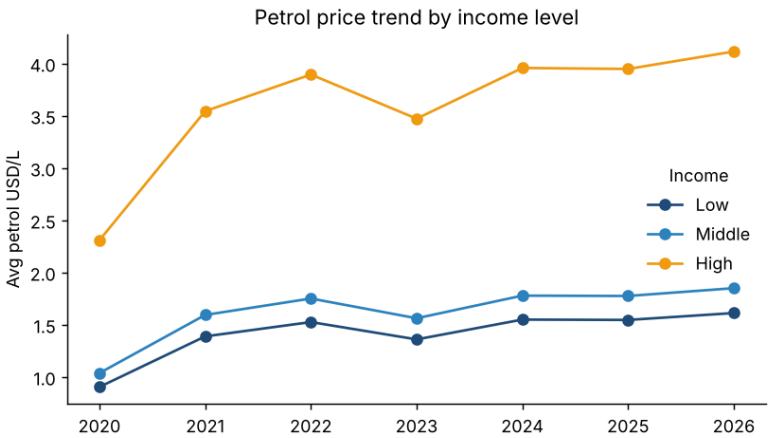


Figure 11: Petrol price trend by income level

7. Dashboard Design

The dashboard has three views, one per persona, sharing global filters for region, income level, subsidy level, start year, end year and fuel type.

View	Contents
1. Policy (Dr. Mitchell)	KPI tiles; average price and tax by region; price by subsidy level; tax versus price scatter by country.
2. Logistics (M. Thompson)	Price trend by year; price versus Brent; 10 cheapest and 10 most expensive countries; volatility ranking.
3. Research (Prof. Rodriguez)	Price and tax by income level; trend by income level; income-by-subsidy heat table with country counts.

The HTML dashboard is included in the repository (dashboard/index.html). The same worksheets can be rebuilt in Tableau using the cleaned CSV, with the world map of average petrol price by country as an additional Tableau view.

8. Recommendations

- **For policy makers:** Review tax and subsidy together. Low-subsidy countries tax fuel at about 37% while others tax at about 15%, so tax reform in a subsidised economy should be paired with subsidy reform. Countries with Very High subsidies (under 0.25 USD/L) carry large fiscal exposure when Brent rises.
- **For logistics and procurement:** Favour fuelling in the Middle East, North Africa and South America where pump prices are lowest, but budget for volatility in Venezuela, Iran and Libya. Prefer diesel where possible, and use Brent as the lead indicator for budget forecasts.
- **For researchers and development agencies:** Analyze affordability by subsidy tier as well as income level, and model each country separately when measuring the effect of Brent on prices.

9. Limitations

- The dataset contains one fuel-price series per country; there is no within-country regional variation or household income data, so affordability is inferred from income group rather than measured.
- Petrol, diesel and LPG are almost perfectly correlated, so they add little independent information.
- Weekly tax percentages fluctuate strongly, so tax conclusions rely on averages.
- Income and subsidy levels are fixed labels per country; they do not change over the period.
- 2026 covers only January to April.

10. Conclusion

Global fuel prices from 2020 to 2026 are shaped by three layers: the world oil price, which moves every country's prices over time; tax policy; and subsidy policy, which separates countries into very different price tiers. A three-view dashboard turns these relationships into

answers for a policy consultant, a logistics manager and a researcher. Future work could add a forecasting model for price, a Tableau map view, and external data such as GDP per capita to measure affordability directly.

11. References

- Dataset: global_fuel_prices_Dataset.xlsx, provided with the SkillWallet capstone project.
- Tableau Public documentation: https://help.tableau.com
- pandas documentation: https://pandas.pydata.org/docs/
- Chart.js documentation: https://www.chartjs.org/docs/